

Yixuan Huang

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Research Experience

Postdoctoral Research Associate with Prof. Tom Silver <i>Princeton University</i>	<i>Princeton, NJ</i> Aug 2025 – Present
Visiting Student Researcher with Prof. Jeannette Bohg <i>Stanford University</i>	<i>Stanford, CA</i> Jan 2024 – Aug 2024
Graduate Research Assistant with Prof. Tucker Hermans <i>University of Utah</i>	<i>Salt Lake City, UT</i> May 2021 – Aug 2025
Undergraduate Research Assistant with Prof. Sicun Gao <i>University of California, San Diego</i>	<i>La Jolla, CA</i> Aug 2019 – Aug 2020

Education

University of Utah, Salt Lake City, UT <i>Ph.D. in Computing: Robotics</i> ◦ GPA: 3.98/4.0 ◦ Advisor: Prof. Tucker Hermans	Aug 2020 - Aug 2025
Northeastern University, Shenyang, Liaoning (China) <i>B.E. in Computer Science and Technology (<u>top</u> student in the department)</i> ◦ GPA: 93.2/100, Rank: 1/278	Sep 2016 - Jun 2020
University of California, San Diego, La Jolla, CA <i>Exchange student</i> ◦ GPA: 3.91/4	Sep 2018 - Jun 2019

Journal Publications

Yixuan Huang, Nichols Crawford Taylor, Adam Conkey, Weiyu Liu, Tucker Hermans. “Latent Space Planning for Multi-Object Manipulation with Environment-Aware Relational Classifiers,” *IEEE Transactions on Robotics (T-RO)* 2024. [Project Website] [Paper]

Conference Publications

Yixuan Huang*, Bowen Li*, Vaibhav Saxena*, Yichao Liang, Utkarsh Aashu Mishra, Liang Ji, Lihan Zha, Jimmy Wu, Nishanth Kumar, Sebastian Scherer, Danfei Xu, Tom Silver. “KinDER: A Physical Reasoning Benchmark for Robot Learning and Planning,” *Robotics: Science and Systems (RSS)*, 2026. [Project Website] [Paper] [Code]

Shuangyu Xie, Kaiyuan Chen, Ziyang Chen, Simeon Adebola, **Yixuan Huang**, Zehan Ma, Tianshuang Qiu, Wentao Yuan, Dhruv Shah, Pannag R Sanketi, Ken Goldberg. “RoboVista: Evaluating Vision Language Models for Diverse Robot Applications,” *Robotics: Science and Systems (RSS)*, 2026.

Lihan Zha, Asher James Hancock, Mingtong Zhang, Tenny Yin, **Yixuan Huang**, Dhruv Shah, Allen Z. Ren, Anirudha Majumdar. “LAP: Language-Action Pre-training Enables Zero-Shot Cross-Embodiment Transfer,” *Robotics: Science and Systems (RSS)*, 2026. [Project Website] [Paper] [Code]

Mingen Li, Houjian Yu, **Yixuan Huang**, Youngjin Hong, Changhyun Choi. “Hierarchical DLO Routing with Reinforcement Learning and Multimodal In-Context VLMs,” *IEEE International Conference on Robotics and Automation (ICRA)* 2026. (**Best Paper Award on Robot Learning – Finalist**) [Project Website] [Paper]

Yixuan Huang, Novella Alvina, Mohanraj Devendran Shanthi, Tucker Hermans. “Fail2Progress: Learning from Real-World Robot Failures with Stein Variational Inference,” *Conference on Robot Learning (CoRL)*, 2025 [Project Website] [Paper]

Yixuan Huang, Christopher Agia, Jimmy Wu, Tucker Hermans, Jeannette Bohg. “Points2Plans: From Point Clouds to Long-Horizon Plans with Composable Relational Dynamics,” *IEEE International Conference on Robotics and*

Automation (ICRA) 2025. [Project Website] [Paper]

Yixuan Huang, Jialin Yuan, Chanh Kim, Pupul Pradhan, Bryan Chen, Li Fuxin, Tucker Hermans. “Out of Sight, Still in Mind: Reasoning and Planning about Unobserved Objects with Video Tracking Enabled Memory Models,” IEEE International Conference on Robotics and Automation (ICRA) 2024. [Project Website] [Paper]

Yixuan Huang, Adam Conkey, Tucker Hermans. “Planning for Multi-Object Manipulation with Graph Neural Network Relational Classifiers,” IEEE International Conference on Robotics and Automation (ICRA) 2023. [Project Website] [Paper]

Yixuan Huang, Michael Bentley, Tucker Hermans, Alan Kuntz. “Toward Learning Context-Dependent Tasks from Demonstration for Tendon-Driven Surgical Robots,” International Symposium on Medical Robotics (ISMR) 2021. **(Best Paper Award – Finalist & Best Student Paper Award – Finalist)** [Paper]

Workshop Publications

Siyeon Kim, Mohanraj Devendran Shanthi, **Yixuan Huang**, Tucker Hermans. “Learning Multimodal Probabilistic Models of Manipulation Skill Effects,” 2025 CoRL Workshop on Learning to Simulate Robot Worlds.

Yixuan Huang, Novella Alvina, Mohanraj Devendran Shanthi, Tucker Hermans. “Fail2Progress: Learning from Failures with Stein Variational Inference for Robot Manipulation,” 2025 RSS Workshop on Out-of-Distribution Generalization in Robotics Towards Reliable Learning-Based Autonomy. **Best Paper Award – Finalist**

Yixuan Huang, Christopher Agia, Jimmy Wu, Tucker Hermans, Jeannette Bohg. “Points2Plans: From Point Clouds to Long-Horizon Plans with Composable Relational Dynamics,” 2024 CoRL Workshop on Learning Effective Abstractions for Planning (LEAP). **Oral Presentation**

Yixuan Huang, Jialin Yuan, Weiyu Liu, Chanh Kim, Li Fuxin, Tucker Hermans. “Latent Space Planning for Unobserved Objects with Environment-Aware Relational Classifiers,” 2023 IROS Workshop Causality for Robotics.

Yixuan Huang. “Reasoning and Planning about Unobserved Objects with Memory Models,” 2023 CoRL Workshop for Neural Representation Learning for Robot Manipulation.

Honors and Awards

2026 ICRA Best Paper Award on Robot Learning – Finalist	May 2026
Rising Star at the Princeton Robotics Inaugural Symposium	April 2026
2025 ICRA Doctoral Consortium	May 2025
2021 ISMR Best Paper Award – Finalist	Nov 2021
2021 ISMR Best Student Paper Award – Finalist	Nov 2021
2021 ISMR NSF Travel Award	Oct 2021
University of Utah School of Computing Department Fellowship	Aug 2020
National Scholarship (top 1%)	Nov 2017 & 2018
Northeastern University Excellent Student (top 1%)	Dec 2017 & 2018
Second Place in National Mathematical Modeling Competition in China	Oct 2017
First Place in Provincial Mathematical Modeling Competition	Oct.2017

Invited Talks

Mar 2026, “Toward Structured and Adaptive Robot Physical Reasoning”, Host: Felix Heide, Princeton University, Princeton, NJ.

Sep 2025, “Planning for Sequential Multi-Object Robot Manipulation with Learned Relational Dynamics Models”, Princeton Postdoctoral Council Seminar Series, Princeton, NJ.

July 2025, “Planning for Sequential Multi-Object Robot Manipulation with Learned Relational Dynamics Models”, Oregon State University, Host: Li Fuxin, Corvallis, OR.

May 2025, “Planning for Sequential Multi-Object Robot Manipulation with Learned Relational Dynamics Models”, University of Texas at Austin, Host: Roberto Martín-Martín, Austin, TX.

Apr 2025, “From Sensor Data to Long-Horizon Plans with Spatial-Temporal Reasoning”, University of Utah Robotics Seminar, Host: Haohan Zhang, Salt Lake City, UT.

Jan 2025, "From Sensor Data to Long-Horizon Plans with Spatial-Temporal Reasoning", University of Utah Graduate Research Symposium, Host: Pratik Soni, Salt Lake City, UT.

Feb 2024, "Reasoning and Planning for Unobserved Objects with Environment-Aware Relational Classifiers and Memory Models", University of Illinois Urbana-Champaign, Host: Katie Driggs-Campbell, Champaign, IL.

Nov 2021, "Toward Learning Context-Dependent Tasks from Demonstration for Tendon-Driven Surgical Robots", University of Utah Robotics Seminar, Host: Jake J Abbott, Salt Lake City, UT.

Teaching

ECE 531/COS 531: Robot Planning Meets Machine Learning <i>Guest Lecture about Learning Models for Robot Planning</i>	Princeton University Oct 2025
CS 6300 Artificial Intelligence <i>Guest Lecture about Reinforcement Learning</i>	University of Utah Nov 2024
CS 4300 Artificial Intelligence <i>Guest Lecture about Imitation Learning</i>	University of Utah April 2022
CS 4300 Artificial Intelligence <i>Teaching Assistant</i>	University of Utah Jan 2022 – May 2022
CS 4300 Artificial Intelligence <i>Teaching Assistant</i>	University of Utah Aug 2022 – Dec 2022

Mentoring

*Mentees who co-authored above listed publications/articles are indicated with *.*

Nichols Crawford Taylor* (Utah Undergrad → Northeastern PhD Student)	Fall 2021 - Spring 2024
Pupul Pradhan* (Utah Master → Evans & Sutherland)	Summer 2023
Novella Alvina* (Utah Master)	Fall 2024 - Summer 2025
John Chen (Utah Undergrad)	Spring 2025-Summer 2025
Bingying Wang (Utah Undergrad)	Spring 2025
Liang Ji* (Princeton Visitor)	Fall 2025 - Present
Joseph Xu (Princeton Master)	Fall 2025 - Present
Ian Henriques (Princeton Master → MIT PHD Student)	Fall 2025 - Spring 2026
Tom Wang (Princeton Undergrad)	Fall 2025
Akash Bhowmick (Princeton Undergrad)	Fall 2025

Reviewing

Conference on Robot Learning (CoRL)	2023 - Present
Robotics: Science and Systems (RSS)	2024 - Present
Robotics and Automation Letters (RA-L)	2024 - Present
International Conference on Robotics and Automation (ICRA)	2023 - Present
International Conference on Intelligent Robots and Systems (IROS)	2024 - Present
International Conference on Learning Representations (ICLR)	2025 - Present
Neural Information Processing Systems (NeurIPS)	2026 - Present
IEEE Transactions on Robotics (T-RO)	2025 - Present
The International Journal of Robotics Research (IJRR)	2026 - Present
IEEE Transactions on Artificial Intelligence	2024

IEEE Transactions on Instrumentation and Measurement	2024
IEEE Transactions on Industrial Electronics	2024
Workshop on Learning Effective Abstractions for Planning (LEAP @ CoRL)	2024, 2025
RSS Workshop on Out-of-Distribution Generalization in Robotics Towards Reliable Learning-Based Autonomy	2025

Skills

Computer Languages: C/C++, MATLAB, Python (TensorFlow, PyTorch), Java, VHDL

Software & Tools: IsaacGym, ROS, Gazebo, PyBullet, HTML, LaTeX

Outreach

Robotics lab tour co-organizer, University of Utah Bridge Program	2022, 2023
Introduction to mobile manipulator, Undergraduate AI Club at Princeton	2025
Introduction to robot programming for 7th-10th-grade students.	2026